its disintegration the radioactive atom is indistinguishable from its stable isotope except for the isotopic mass difference. When the radioactive disintegration of the atom has been observed ("counted"), the decay product is normally a different element and its subsequent chemical behavior is usually of no interest. If the disintegration rate is very high, the energy released by the radioactive decay can cause observable secondary radiolytic effects (Ch. 8). However, in well-designed tracer experiments the level of radioactivity is high enough to provide accurate data but normally small enough not to produce noticeable chemical effects.

While the radioactivity of the tracers is assumed not to affect the chemical systems, the parent-daughter relationship of radioactive nuclides needs special consideration. For example, since strontium and yttrium are not chemically identical, a gross β-count of strontium samples including ^{90}Sr may include an unknown fraction of ^{90}Y activity present from ^{90}Sr decay because of the relationship

$$^{90}\mathrm{Sr}(\beta^-, t_{½}\ 28.5\ \mathrm{y})\ ^{90}\mathrm{Y}\ (\beta^-\gamma, t_{½}\ 2.671\ \mathrm{d})\ ^{90}\mathrm{Zr}(\mathrm{stable}) \qquad (18.1)$$

Beta-absorption, and β- or γ-scintillation techniques which use energy discrimination, are frequently useful in such parent-daughter cases. If equilibrium is rapidly established between the parent and daughter activities it is usually simpler to count the samples after sufficient time for this to occur, the contribution to the observed count rate by the daughter is then proportional to the amount of mother in the sample. In the case of ^{90}Sr—^{90}Y, radioactive equilibrium is established in about 25 d. If ^{137}Cs ($\beta^-\gamma$, $t_{½}$ 30.0 y) is being used to study cesium chemistry it is necessary to wait only 15−20 min after sampling until counting as the daughter ^{137m}Ba (IT, $t_{½}$ 2.55 min) reaches an equilibrium level within that time. Since the ratio of the ^{137m}Ba and the ^{137}Cs activity is the same in all samples at equilibrium, the total count rate before and after a chemical step is then a true measure of the behavior of cesium alone. If radioactive equilibrium is not re-established in a convenient time, it may be necessary to either discriminate against the activity not involved in the chemical system, to take into account its contributions to the net count rate, or to remove it immediately before counting.

It may be necessary or expedient to use a radioactive nuclide which can undergo significant decay during the chemical investigation. In these cases, in order to compare results at different points in the process, it is necessary to correct all counts to the same time (usually starting time of the experiment).

18.2. CHEMISTRY OF TRACE CONCENTRATIONS

Consider a sample containing a pure radionuclide with a disintegration rate of 10^7 dpm. For a $t_{½}$ of 1 h the number of atoms is (§5.11) 8.7×10^8; for a $t_{½}$ of 1 y it is 7.6×10^{12}. If such a sample is dissolved in one liter of solution, the respective concentrations would be 1.4×10^{-15} M and 1.3×10^{-11} M. At such concentrations the chemical behaviour may

be quite different from what it is at higher concentrations. Addition of macroscopic amounts (e.g. at the gram level) of non-radioactive isotopic) atoms of the element results in concentrations of 10^{-3} to 10^{-1} M. The non-radioactive component is called a *carrier* as it "carries" the radioactive and ensures normal chemical behaviour. Many applications of radiotracers involve mixing the tracer atoms with a much larger amount of nonradioactive isotopic atoms prior to use.

If a radionuclide is to follow the chemical properties of an isotopic carrier it is necessary that the radionuclide and the carrier undergo *isotopic exchange*. If it is not known a priori that such exchange takes place between two compounds with a common element this must be determined by experimentation before it can be assumed that the tracer and the carrier would act similarly in a chemical system. This consideration must be particularly borne in mind if the radioactive tracer and the inert carrier are in different oxidation states when mixed.

In the remainder of this paragraph we discuss the behaviour of trace level concentrations as it is desirable in some applications to use very low concentrations of radiotracers with no carrier.

18.2.1. Adsorption

Solutes in contact with surfaces have a tendency to be adsorbed on the surface. In order to cover the glass surface of a one liter vessel with a monomolecular layer of a hydrated cation only 10^{-7}—10^{-8} moles are required. As indicated in the previous paragraph, the amount of radionuclide in the solution may be less than this and, in principle, all the radioactive atoms could be adsorbed on the walls of the vessel. The *Paneth and Fajans rule* for tracer adsorption states that: "a micro component is adsorbed on a solid macro component or precipitated together with it if it forms an insoluble compound with a counter ion of the macro component".

The amount of radionuclide that is adsorbed on the walls of the container depends on the concentration, on the chemical state of the radionuclide and on the nature of the container material. One of these properties of the material is the so called point of zero charge (pzc). This is the pH where the surface charge is zero. At a pH below this point the surface is positively charged and anios will adsorb. At pH above this point the surface will be negativelay charged and thus only cat ions will be adsorbed. Figure 18.1 shows the variation of the adsorption of thorium on the walls of glass and polyethylene containers as a function of concentration and pH. In case (a) the sorption time is that of pipetting (of an aqueous Th-complex), in case (b) it is that for equilibrium. The variation of adsorption with pH reflects the adsorption of various hydrolytic species formed by thorium as the pH is increased. Curve (a) shows that sorption can be neglected at concentrations $>10^{-4}$ M in this system.

In many cases in nature the pH of the solution in contact with the sold phase is higher than the point of zero charge for that surface. In general adsorption of cations increases

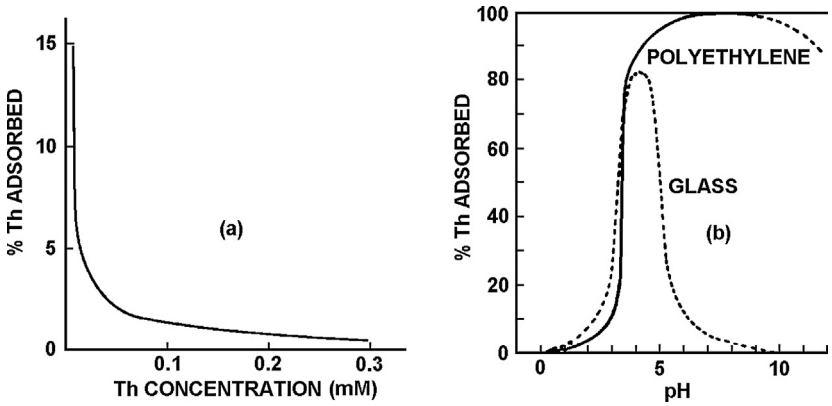


Figure 18.1 Adsorption of Th (a) in a 10 ml pipette at different Th concentrations, (b) from 2×10^{-8} M $Th(ClO_4)_4$ solution on different surfaces and pH's. *(From Rydberg and Rydberg.)*

with ionic charge in the order $M^+ < M^{2+} < M^{3+} < M^{4+}$. The importance of the nature of the surface is obvious in Figure 18.1.b. Adsorption of Pm(III) ions have been shown to increase in the order platinum < silver < stainless steel < polyvinyl chloride. Addition of isotopic carrier dilutes the radiotracer and a smaller fraction of tracer is adsorbed (Fig. 18.1.a). Unfortunately, such isotopic dilution results in a decrease in the specific activity of the trace element, which can be disadvantageous in certain types of experiments. In some cases it is possible to avoid decreasing the specific activity by adding macro amounts of a nonisotopic element which is easily adsorbed and may block the available surfaces from adsorbing the tracer.

In addition to adsorption on the walls of the container, radioactive species frequently adsorb on precipitates present in the system. The nature of the precipitate as well as its mode of precipitation are major factors in the amount of adsorption. If silver iodide is precipitated in an excess of silver ion the precipitate has a positive surface layer due to the excess concentration of silver ions on the surface. By contrast if the precipitation occurs in excess iodide, there is a negative surface charge due to the excess iodide on the surface. When trace amounts of radioactive lead ions are added to a suspension of two such precipitates in water, the precipitate with the negative surface charge adsorbs >70% of the tracer lead ions from the solution, while the precipitate with the positive surface charge adsorbs <5%. The amount of adsorption increases with the ionic charge of the radioactive tracer, e.g. it has been found that with a precipitate of Ag_2S about 7% of Ra^{2+}, 75% of Ac^{3+} and 100% of Th^{4+} is adsorbed.

The adsorption properties of trace elements have been used to advantage for the isolation of the trace elements as well as for the separation of different trace elements with different adsorption properties.

18.2.2. Radiocolloids

Radioactive tracers adsorb not only on solid container surfaces and precipitates but on any kind of solid material suspended or in contact with the solution. Dust, cellulose fibers, glass fragments, organic materials, etc., are examples of substances that readily adsorb radioactive tracers from solution. If the solution contains large molecules as, for example, polymeric metal hydrolysis products, these also tend to adsorb trace elements. In addition to sorption, the presence of such material in the solution can lead to the phenomenon of radiocolloid formation, which is the attachment of radionuclides to semi-colloidal aggregates in solution. If the solution is kept at sufficiently low pH and extremely free from foreign particles, sorption and radiocolloid formation are usually avoided as major problems.

18.2.3. Equilibrium reactions

The low concentration of radioactive tracers can lead to the formation of solute species that are not observed at equilibrium with macro amounts. For example, the hydrolysis of uranyl ions corresponds to the equilibrium

$$m\,UO_2^{2+} + p\,H_2O \;\rightleftarrows\; (UO_2)_m(OH)_p^{2m-p} + p\,H^+ \tag{18.2}$$

With macro concentrations of uranium this equilibrium is shifted to the right with the observation of polymers with properties rather different than that of the uranyl ion. At a uranium concentration of approximately 0.001 M, more than 50% of the uranium is polymerized at pH 6, while for uranium concentrations less than 10^{-6} M the polymerization is negligible. This condition can be used to advantage: trace metal concentrations can be used if one wishes to study the properties of a metal ion at relatively high pH's without interference of polymerization reactions. However, one then must be aware of the adsorption risks.

An additional complication can arise in solution if the radioactive species in trace amounts react with trace concentrations of impurities. For example, in an investigation of the properties of pentavalent protactinium, Pa(V), it was found that the protactinium was extracted into pure xylene from 1 M $HClO_4$ solutions. Further experimentation showed that this extraction was due to the presence in the xylene of organic impurities at concentrations below the detectable limit of 0.01%. Support for this interpretation was provided when the solution was made 10^{-4} M in thorium, which was expected to form complexes with the probable impurity, thereby preventing the reaction with the protactinium. In fact, no protactinium was extracted into xylene from this solution. The thorium in this case acts as a *hold-back carrier.*

18.2.4. Precipitation and crystallization

Due to the low concentration of radioactive tracers in solution the solubility product for an "insoluble" salt is not always exceeded upon the addition of macro concentrations of a

counter ion. Let us as example take the insoluble lanthanum hydroxides. The solubility product for the reaction $La(OH)_3$ (s) $\rightleftarrows$ $La^{3+} + 3\ OH^-$ is $K_{s0} \approx 10^{-19}$; in 1 mM NaOH the concentration of La^{3+} in the solution is only 10^{-10} M in equilibrium with the $La(OH)_3$ precipitate. If 100 MBq (2.7 mCi) ^{140}La, obtained as a fission product or by milking from ^{140}Ba (Table 5.1), is dissolved in 1 l, the solution will have $[La^{3+}] \approx 3.5 \times 10^{-11}$ M. In this case the solubility product is not exceeded in 1 mM NaOH. To precipitate the $La(OH)_3$ quantitatively the NaOH concentration must be raised to 10−100 mM; however, at these concentrations the precipitate is formed as a colloid in solution and even upon centrifugation the amount of precipitate is so small as to be unweighable by present techniques. With the addition of a *carrier* for ^{140}La, the precipitation can be carried out without any difficulty.

It is possible to remove ions at tracer level concentrations from solutions by precipitation using adsorption or coprecipitation. *Coprecipitation* occurs if the compound of the tracer and the oppositely charged ion of the precipitate is isomorphous with the precipitate. In these cases the active ion may be included in the crystal lattice of the precipitate at a lattice point, particularly if the tracer ion is close in size to the ion which it displaces. However, at trace level concentrations exceptions are found to this requirement of similarity in size as well as to the requirement of isomorphism. When the distribution of the tracer is found to be uniform throughout the precipitate it can be described by the *Berthelot-Nernst homogeneous distribution law* which is expressed as

$$x/y = D'(a - x)/(b - y) \tag{18.3}$$

where x and y are the amounts of A^{z+} and B^{z+} in the precipitate, a and b are the initial amounts of these ions, and D' is the "distribution coefficient". A more "true" distribution constant (D = concentration of tracer in solid/concentration of tracer in solution) can be obtained by using a conversion factor, e.g. C = gram solute per ml of saturated carrier solution divided by the density of the solid

$$D' = D\ C \tag{18.4}$$

The entire precipitate is in equilibrium with solution in this system.

If only the freshly forming surface of the growing crystal is in equilibrium with the solution phase, a nonuniform distribution is observed. In these cases the system is described by a logarithmic (according to Doerner and Hoskins) distribution law, which has the form

$$\ln[a/(a - x)] = \lambda' \ln[b/(b - y)] \tag{18.5}$$

where λ' is the *logarithmic distribution coefficient*, a constant characteristic of the system (Fig. 18.2).

The importance of isomorphism can be illustrated by the coprecipitation of Ra^{2+} in trace quantities with Sr^{2+} in strontium nitrate. If the precipitation is carried out at 34°C, the radium coprecipitates since at this temperature the strontium precipitates as $Sr(NO_3)_2$

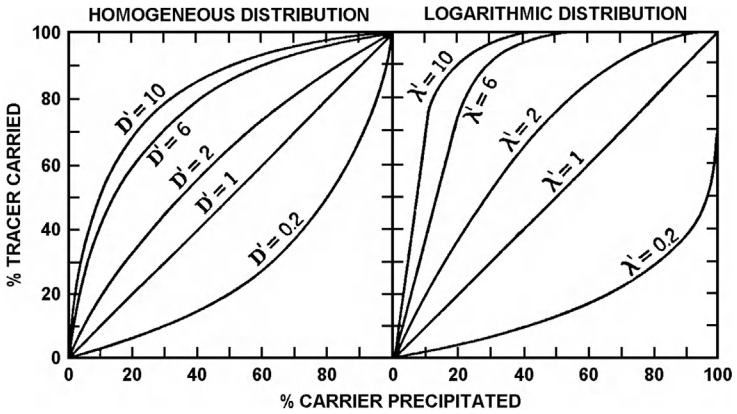


Figure 18.2 Efficiency with which the tracer is carried for various values of the distribution coefficients D' and λ'. *(From Wahl and Bonner.)*

with which radium nitrate is isomorphous. However, if the precipitation occurs at 4°C, the strontium crystallizes as $Sr(NO_3)_2{\cdot}4H_2O$ and is no longer isomorphous with $Ra(NO_3)_2$. Due to the lack of isomorphism the radium is not coprecipitated at 4°C.

18.2.5. Electrochemical properties

For the redox equilibrium

$$M^{z+} + ne^- \rightleftarrows M^{z-n} \; (n \geq 1)$$

where M^{z+} and M^{z-n} are the oxidized and reduced states of a chemical species, the *Nernst equation* is valid, i.e.

$$E = E^0 - \{\mathbf{R}T/(n\mathbf{F})\} \ln\left(\left[M^{z-n}\right]/\left[M^{z+}\right]\right) \tag{18.6}$$

In this equation E^0 is the potential for a standard state of 1 M concentration at NTP; the species in brackets relates to the chemical activities in the particular solution phase. This relationship indicates that the redox potential E of a solution is independent of the total concentration of the species and depends only on the ratio of the oxidized and reduced forms. This has been confirmed since concentrations of trace amounts of ions show the same redox behavior as macro concentrations. Reduction and oxidation reactions can, therefore, be carried out in solutions with trace amounts of radioactive species.

Electrolysis of solutions can be used for electrodeposition of a trace metal on an electrode. The selectivity and efficiency which would be present for electrolytic deposition of macro amounts of ions at a controlled potential is not present, however, for trace amounts. The activity of trace amounts of the species is an unknown quantity even if the concentration is known, since the activity coefficient is dependent upon the behavior of

the mixed electrolyte system. Moreover, the concentration of the tracer in solution may not be known accurately since there is always the possibility of some loss through adsorption, complex formation with impurities, etc. Nevertheless, despite these uncertainties it has been found that the Nernst equation can be used, with some caution, for calculating the conditions necessary for electrolytic deposition of trace metals.

It is also possible to precipitate insoluble species on electrodes. For example, if a fluorosilicate solution is electrolyzed, thereby freeing a high concentration of fluoride ion at the electrode, a thin uniform layer of UF_4, can be deposited. Similarly, trace amounts of elements which form insoluble hydroxides can be deposited from solutions in which water is being electrolyzed as a region of extremely high pH is present at the cathode.

18.2.6. Tracer separation methods

All the analytical techniques used in conventional chemistry may be used for the separation and isolation of radioactive elements and compounds in macro or trace concentrations. The precipitation method was amply demonstrated by the early radio-chemists M. Curie, Debierne, Rutherford, Hahn, etc., for the separation, concentration and identification of the naturally occurring radioactive elements. However, in §§18.2.1—18.2.4 we have pointed out the many pitfalls in working with tracer concentrations in solutions containing precipitates, etc, as well as in the use of electrochemical methods (§18.2.5).

Normally these separation methods require the addition of a macro amount of isotopic carrier. However, in some cases analytical procedures are available for separation and isolation of carrier free radiotracer concentrations. *Solvent extraction* (see §18.4.3 and App. A), and various forms of *partition chromatography* (§18.4.1), methods have been found to be particularly advantageous in this connection since they are selective, simple, and fast.

Liquid-liquid (or *solvent*) *extraction* is a technique for selectively transferring a species between an aqueous solution and an organic phase (e.g. kerosene, benzene, chloroform, etc) by equilibrating the aqueous phase with an organic solvent. Usually the organic phase contains a reagent A (*extractant*) which forms a neutral compound MA_N with the species M to be transferred between the phases. The number of extractants applied are numerous and the literature must be consulted to determine the most suitable ones for the system of interest; typical extractants are organophosphates, amines and metal chelating agents (usually weak organic acids). The fraction extracted at equal phase volumes is

$$E\,\% = 100\,D/(D+1) \tag{18.7}$$

where D is the *distribution ratio* of the radioactivity between the two phases

$$D = R_{org}/R_{aq} = \psi_{org}A_{org}/\psi_{aq}A_{aq} \approx N_{org}/N_{aq} = [M]_{org}/[M]_{aq} \tag{18.8}$$

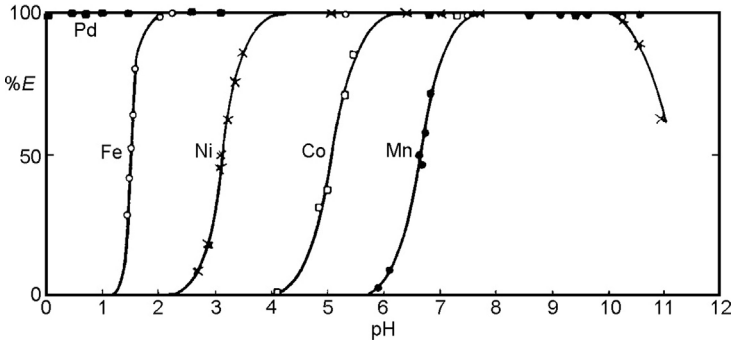


Figure 18.3 Effect of pH on the extraction of some divalent metals from aqueous solution into chloroform by 0.01M 8-hydroxyquinoline (oxine).

The last equality requires that the radioactive measurements R are carried out on equal phase volumes (eventually evaporated to dryness) and that $\psi_{org} = \psi_{aq}$, a requirement easily met by proper choice of radiometric equipment. Thus the D-value directly reflects the concentration ratio of the radioactive species. Figure 18.3 shows as an example how the extraction of a number of metals from an aqueous into an organic solution varies with pH. Such curves are used to select optimal separation conditions: in the Figure at pH <3.5, Pd, Fe and Ni are extracted to 100% into chloroform, while all the Co and Mn stays in the aqueous phase, thus facilitating an easy separation of these two groups of metals.

This technique has a number of applications, i.e.

- on a large scale for the production of valuable metals, such as U, described in §14.2.4,
- for reprocessing spent nuclear fuels as described in Ch. 21,
- at trace metal concentrations for determining equilibrium constants as described in §18.4.3,
- for separation and identification of short lived radionuclides as described in §17.7.

Solid organic resin *ion exchangers* consist of organic polymeric networks containing basic or acidic groups attached to the organic framework. Analogous to (18.8) a distribution ratio D_{iex} is defined as

$$D_{iex} = [M]_{resin}/[M]_{sol} \tag{18.9}$$

for the distribution of a (e.g radioactive) metal between the solid resin and the aqueous solution. D_{iex} depends on resin properties and on solution parameters such as the nature of the metal ion, ionic strength of the solution, temperature, etc. The basic equilibria are discussed in §18.4.3. Because the sorption in the resin phase increases with the valency of the cation, multivalent ions are absorbed more strongly (i.e. have larger D_{iex} values) than divalent or monovalent ions. Most commonly ion exchange columns are used for metal ion separations. In this case, the radioactivity is sorbed in the top layer of a column of the

wet ion exchange resin. Following sorption (e.g. of M^+), the metal is eluted from the resin bed by passage of a solution (eluant) through the column. The eluant may contain a complexing anion or another metal ion (e.g. M^{3+}) which displaces M^+ through competition with it for positions on the resin. The metals are eluted by complexing agents in an order depending on their complex formation properties, as e.g. is illustrated for the lanthanides and actinides in Chapter 14 (Fig. 14.7).

In principle, *liquid partition chromatography* (LPC) is a liquid-liquid extraction where one of the liquid phases is stationary and attached to a supporting material, and the other liquid phase is mobile. It can be carried out with either the aqueous or the organic phase stationary; in the latter case the technique is referred to as reversed phase LPC. The aqueous phase can be made stationary by adsorption on silica gel, cellulose powder, etc. In order to make the organic phase stationary, beads (usually 50–200 μm) of PVC, teflon, Kel-F, etc., are being used.

Reversed phase LPC has been useful in radiochemistry for separating individual elements, e.g. lanthanides or actinides. It has also been used for separation of macro amounts of actinides. Instead of using columns in partition chromatography, a sheet of paper may be used to hold the stationary phase (*paper chromatography*) or an adsorbent coated on a glass plate (*thin-layer chromatography*). This technique has an advantage over column separations because the positions of the radioactive species are easily identified on the sheet, either simply by autoradiography (§18.4.4) or by scanning instruments. Paper chromatography is further described in an example in §18.4.1 and illustrated by Figures 18.7 and 18.8.

18.3. ANALYTICAL CHEMISTRY

18.3.1. Radiometric analysis

The term radiometric analysis is often used in a broad sense to include all methods of determination of concentrations using radioactive tracers. In a more restricted sense it refers to a specific analytical method which is based on a two-phase titration in the presence of a radioactive isotope. The endpoint of the titration is indicated by the disappearance of the radioisotope from one of the phases. Figure 18.4 illustrates two cases, (a) the determination of Ag^+ in a solution by titration with NaI solution containing $^{129}I^-$ ($\beta^-\gamma$ $t_{½}$ 1.57×10^7 y), and (b) the determination of Fe^{2+} in an aqueous solution, to which trace amounts of radioactive $^{55}Fe^{2+}$ (EC $t_{½}$ 2.73 y) has been added. In case (a) the AgI precipitate is radioactive but the solution has little radioactivity until all the Ag^+ has been precipitated. The activity of the solution is measured by a liquid flow GM-detector (Ch. 9). In the latter case (b) a two-phase liquid-liquid analytical technique is used (§18.2.6); the titrant contains a substance (oxine) which extracts Fe(II) from the aqueous to the chloroform phase. The radioactivity of the organic phase is followed by liquid scintillation (sampling) to determine the end point of the titration.